

Exponent Laws with Variable Bases*Question Bank - Additional 3.T1*

Show your work. Circle your final answer.

For questions 1 through 6, write an equivalent expression as a single monomial. Assume no variable equals 0.

1. $y^6 \cdot y^7$

2. $(k^3)^8$

3. $(5x^2)^3$

4. $\frac{b^{11}}{b^4}$

5. $(-3a^5)(-4a^2)$

6. $\frac{18x^6y^4}{6x^5y}$

For questions 7 and 8, apply more than one law. Write each answer as a single monomial.

7. Write an equivalent expression.

a. $(3a^4b^2)(6ab^3)$

b. $(2p^3q^5)^3$

c. $(xy^3)^2 \cdot x^4y$

8. Write an equivalent expression.

a. $\frac{(2x^3y)^4}{8x^7y}$

b. $\frac{m^5n^6}{m^2n^6}$

c. $\frac{(5c^2)^2}{5c^3}$

Working backwards.

9. Find the value of n that makes each statement true.

a. $a^n \cdot a^5 = a^{12}$

b. $(x^3)^n = x^{18}$

c. $\frac{y^n}{y^4} = y^6$

10. A rectangle has length $9a^5$ and width $3a^2$.

a. Write its area as a single monomial.

b. A square has side length $7b^3$. Write its area as a single monomial.

11. Which expression is equivalent to $(4y^2)^3$?

- ☐ A. $12y^6$
- ☐ B. $4y^6$
- ☐ C. $64y^6$
- ☐ D. $64y^5$

12. Select ALL of the expressions that are equivalent to m^{10} .

☐ A. $m^3 \cdot m^7$

☐ B. $(m^2)^5$

☐ C. $m^2 \cdot m^5$

☐ D. $\frac{m^{20}}{m^2}$

☐ E. $\frac{m^{14}}{m^4}$

☐ F. $(m^5)^2$